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Energy-Efficient Predictive RIS Beamforming in Multi-Satellite LEO Networks Using Twin-Delayed Deep Reinforcement Learning

Muhammad Khalil, *Member, IEEE*, Ke Wang, *Senior, IEEE*, and Jinho Choi, *Fellow, IEEE*

***Abstract*—This paper presents an energy-efficient downlink control framework for multi-satellite low-Earth orbit (LEO) networks assisted by terrestrial reconfigurable intelligent surfaces (RIS). The proposed method integrates three key components: (i) an extended Kalman filter (EKF) for mobility and channel prediction; (ii) a single-step affine calibration process that constrains the predicted signal-to-noise ratio (SNR) to a prescribed normalized mean-square error (NMSE); and (iii) a twin-delayed deep deterministic policy gradient (TD3) controller that dynamically adjusts the RIS phase configurations in real time. An analytical model of energy efficiency (EE), accounting for satellite transmit power, RIS circuit power, and processing overhead, has been developed and employed both as a benchmarking reference and as the optimization target for the learning process.**

Simulation results across varied mobility conditions, satellite multiplicities, and RIS configurations show that the EKF-calibration loop maintains the SNR-prediction NMSE at or below 8×10^{-3}, enabling stable predictive control. Increasing the RIS aperture produces the expected ≈ 6 dB SNR improvement with diminishing marginal EE gains. for example, doubling the aperture from 64 to 128 elements improves EE by approximately 59%; from 128 to 256 elements by 29%; and from 256 to 512 elements by 4.5%. With TD3-optimized phases, the achieved EE attains $93{-}97\%$ of the analytical upper bound ($0.5 - 3.4\,\mathrm{bit\,J^{-1}}$), while per-satellite transmit-power savings exceed 50% in typical regimes, and network-wide savings reach up to 12% compared with non-predictive baselines. For moderate satellite diversity with large RIS apertures, the incremental gain from learning-based control is up to 0.38 bits J^{-1}.

These findings demonstrate that the integration of EKF-based motion prediction, NMSE-constrained SNR calibration, and learning-based RIS optimization yields a scalable and near-optimal solution for energy-efficient downlink operations in LEO satellite systems.

***Index Terms*—Mobility prediction, RIS, multi-satellite communication, energy efficiency, extended Kalman filter.**

I. Introduction

Integrating low-Earth orbit (LEO) satellite networks with terrestrial 6G is expected to enable global, high-speed connectivity [1]. LEO constellations provide low latency and wide-area access, but they also introduce challenges that are less severe in terrestrial systems, including fast channel variations, large Doppler shifts, and short visibility windows caused by high satellite mobility [2]. In addition, long propagation distances and blockage lead to severe path loss and intermittent links [3], while tight power and payload budgets constrain robust transmission strategies. These factors make seamless LEO-6G integration technically challenging.

Reconfigurable intelligent surfaces (RIS) enable programmable manipulation of the wireless propagation environment through controllable phase shifts [4]. In LEO systems, RIS-assisted reflection enhances effective channel gain and link reliability, which is particularly beneficial under high mobility and severe path loss [5], [6]. Prior works have investigated RIS-assisted satellite networks from different perspectives. For instance, [7] studies joint ground-station beamforming and RIS optimization for sum-rate maximization, while [8] considers joint passive beamforming and RIS orientation in LEO downlinks. Energy-efficiency optimization is addressed in [6], and RIS-assisted THz LEO links are explored in [9]. Surveys further highlight RIS as a key enabler for 6G non-terrestrial networks due to its gains in coverage, interference mitigation, and spectral efficiency [2]. **More recent studies examine robust RIS/metasurface designs under imperfect geometry and jamming. For example, [10] develops a multi-functional RIS framework for aerial–ground networks under uncertain jammer states and CSI, while [11] proposes a dual-polarized stacked metasurface transceiver with robust optimization under imperfect angular CSI. However, these works do not address joint RIS–satellite–user geometry or energy-efficient RIS control in dynamic multi-satellite LEO downlinks, which is the focus of this paper.**

While the above studies have made important progress, most focus on single-satellite systems or rely on static, model-based RIS optimization. In particular, a fixed satellite link is typically assumed, or offline optimization problems are usually solved for given channel states [12]. However, such approaches lack the agility to handle the dynamic multi-satellite environment of LEO mega-constellations [13]. With multiple LEO satellites in motion, the propagation geometry and optimal RIS configuration change rapidly over time [14]. To the best of our knowledge, existing methods offer no scalable mechanism for coordinating RIS phase adjustments across multiple LEO satellites, nor can they adapt in real time to the network's rapidly changing dynamics. Furthermore, there is a notable absence of adaptive and scalable control frameworks that explicitly address EE in RIS-assisted multi-satellite LEO networks. To date, there is no comprehensive solution capable of jointly optimizing RIS configuration and energy consumption under the practical constraints of dynamic,

The authors Khalil and Wang are with the School of Engineering, RMIT University, Melbourne, Australia. Emails: muhammad.khalil@rmit.edu.au, and ke.wang@rmit.edu.au. The author Choi with the University of Adelaide, Adelaide, Australia. Email: jinho.choi@adelaide.edu.au

multi-satellite environments. Accordingly, the development of such frameworks remains an open and significant research challenge.

To address this gap, our work proposes a novel deep reinforcement learning (DRL)-based framework for the phase control in a multi-satellite LEO network. DRL techniques are well-suited for complex, dynamic optimization problems, as they can learn efficient policies through interaction with the environment without requiring an explicit model of the channel dynamics [15]. In fact, very recent research has begun to apply DRL for resource optimization in non-terrestrial and RIS-assisted networks [16]. For example, multi-agent DRL has been utilized for spectrum allocation in cognitive satellite-terrestrial systems [17], and other works leverage DRL to adjust UAV-mounted RIS and scheduling policies in LEO communications [18]. Moreover, state prediction methods, such as Kalman filtering, have been explored to assist DRL in managing mobility. For instance, in [19] a Kalman filter-based user tracking algorithm has been developed in an RIS-empowered LEO network to proactively predict channel variations due to motion. Inspired by the increasing need for intelligent, adaptive satellite communications, this work integrates an extended Kalman filter (EKF)-based mobility prediction with a DRL controller to achieve real-time, mobility-aware RIS phase optimization in multi-LEO satellite networks. By uniting model-driven channel prediction with data-driven RIS control, we address the fundamental challenge of maximizing EE under dynamic, high-mobility conditions. Simulation results demonstrate that the proposed approach consistently achieves 97% of the analytical energy-efficiency upper bound and can reduce total transmit power by up to 12% across a wide range of satellite and RIS configurations.

The primary contributions of this paper are threefold. First, we develop an analytic EE optimization model tailored to RIS-assisted multi-satellite downlink scenarios and embed this closed-form EE expression as the reward objective within a Twin Delayed Deep Deterministic Policy Gradient (TD3) agent. This approach ensures that the learned DRL policy is physically grounded and converges toward the globally optimal solutions predicted by theory.

Second, we design a mobility-aware RIS control method that leverages an EKF-based channel prediction to proactively adapt the RIS phases in response to user and satellite motion. The EKF continuously estimates and predicts the time-varying channel state, including geometric path loss and Doppler shifts. This enables the TD3 agent to anticipate channel fluctuations and maintain robust coherent beamforming gains despite rapid network dynamics.

Third, we establish the scalability of the proposed framework with respect to both the number of RIS elements and the number of cooperating satellites . The joint EKF–TD3 method is designed to efficiently coordinate RIS phase control across a wide range of system sizes, making it suitable for large-scale dynamic LEO satellite networks. This scalability and effectiveness are demonstrated through comprehensive simulations, which confirm that our framework consistently outperforms static and random-phase baselines in terms of EE and throughput, particularly under challenging high-mobility multi-satellite conditions.

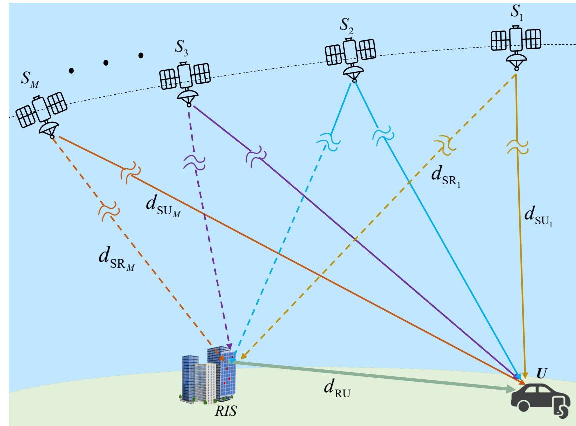

Fig. 1. Geometry of the RIS-assisted multi-satellite LEO downlink system.

By enabling real-time, DRL-based RIS control for multi-LEO constellations with integrated mobility prediction, this work advances beyond the static or single-satellite conditions of previous studies. The framework combines analytical modeling with advanced learning techniques to address both EE and dynamic adaptation in RIS-assisted satellite communications.

The remainder of this paper is organized as follows. Section II presents the multi-satellite LEO downlink geometry and the associated RIS-assisted channel model, and summarizes the key modeling assumptions. Section III develops the EKF-based mobility/state-space representation and the resulting look-ahead channel prediction at the planning horizon. Section IV formulates the real-time RIS phase-control problem as a continuous-state, continuous-action MDP and introduces the TD3-based controller. Section V derives the analytical energy-efficiency metric and shows how it is embedded as the reinforcement-learning reward. Numerical results and discussion are provided in Section VI, and Section VII concludes the paper.

II. User Motion Model

We consider a downlink communication scenario in which a constellation of M LEO satellites, denoted as $S_1, S_2, \ldots, S_M$, cooperates with a fixed, ground-based RIS to serve a mobile user U, as shown in Fig. 1. Each satellite S_m for $m = 1, 2, ..., M$, is located at Cartesian coordinates at time t as follows:

$$\mathbf{r}_{s,m}(t) = \begin{bmatrix} x_{s,m}(t), & y_{s,m}(t), & z_{s,m}(t) \end{bmatrix}^{\mathrm{T}}. \tag{1}$$

The mobile user'geodetic coordinates $(\mathrm{lat}_u, \mathrm{lon}_u)$ are projected onto a local tangent (east-north) plane, resulting in a 2D vector:

$$\mathbf{p}(t) = \begin{bmatrix} x(t), & y(t) \end{bmatrix}^{\mathrm{T}} \in \mathbb{R}^2 \tag{2}$$

The RIS is located at a fixed position on the Earth's surface, denoted by geographic coordinates $(\mathrm{lat}_{\mathrm{RIS}}, \mathrm{lon}_{\mathrm{RIS}})$, which correspond to a Cartesian position vector

$$\mathbf{r}_{\mathrm{RIS}} = \begin{bmatrix} x_{RIS}, & y_{RIS}, & z_{RIS} \end{bmatrix}^{\mathrm{T}}. \tag{3}$$

Each satellite transmits a continuous pilot waveform. We assume a clear line-of-sight (LoS) propagation from each satellite to both the RIS and the user. Accordingly, the signal received by the user is composed of the direct path from each satellite and a single one-bounce reflected path via the RIS. The Euclidean distance of the direct link is then given by

$$d_{\mathrm{SU_m}}(t) = \left\|\mathbf{r}_{s,m}(t) - \mathbf{p}(t)\right\|,$$

The two-hop reflected route is characterized by the distance separation between the satellite and the RIS.

$$d_{\mathrm{SR_m}}(t) = \left\|\mathbf{r}_{s,m} - \mathrm{r}_{\mathrm{RIS}}\right\|,$$

followed by the geodetic distance from the RIS to the user

$$d_{\mathrm{RU}} = \mathrm{VincentyDistance}(\mathrm{lat}_{\mathrm{RIS}}, \mathrm{lon}_{\mathrm{RIS}}, \mathrm{lat}_u, \mathrm{lon}_u) \tag{4}$$

where the Vincenty distance is calculated using the Vincenty formula [20].

To characterize signal attenuation and delay, we employ the free-space path loss model and the definitions of propagation delay. For any link of length d , the power loss is $L(d) = \left(\frac{4\pi f_c d}{c}\right)^2$, and the propagation delay is $\tau = d/c$, where f_c denotes the carrier frequency and c is the speed of light. Hence, at time t we define path losses as follows:

$$\underbrace{L_{\mathrm{SU}_m}(t) = L(d_{\mathrm{SU}_m}(t))}_{\text{LoS path loss}},\ \underbrace{L_{\mathrm{SR}_m} = L(d_{\mathrm{SR}_m}(t))}_{S_m\text{-to-RIS path loss}},$$
$$\underbrace{L_{\mathrm{RU}}(t) = L(d_{\mathrm{RU}}(t))}_{u\text{-to-RIS pathloss}}$$

and the associated delays become

$$\tau_{\mathrm{SU}_m}(t) = \frac{d_{\mathrm{su}_m}(t)}{c},\ \tau_{\mathrm{SR}_m}(t) = \frac{d_{\mathrm{SR}_m}(t)}{c},\ \tau_{\mathrm{RU}}(t) = \frac{d_{\mathrm{RU}}(t)}{c}. \tag{5}$$

It is important to note that while the propagation delays associated with the direct and RIS-reflected paths are explicitly incorporated into the channel phase expressions, their differential effects on symbol timing are not modeled. Specifically, we assume that the resulting delay spread is sufficiently small compared to the symbol duration, so that inter-symbol interference (ISI) can be safely neglected. This assumption is well justified in typical LEO configurations with moderate data rates and compact RIS-user geometries. For scenarios involving very high symbol rates or large RIS-user separations, additional measures, such as guard intervals or equalization, may be required to mitigate ISI, which are beyond the scope of this work.

The RIS consists of N passive reflecting elements, each capable of imposing a phase shift with a unit modulus. The collective phase configuration is represented by the diagonal matrix

$$\mathbf{\Phi}(t) = \mathrm{diag}\left(e^{j\phi_1(t)}, \ldots, e^{j\phi_N(t)}\right),\ \phi_n \in [0, 2\pi). \tag{6}$$

This matrix represents the instantaneous configuration of the RIS, and its selection will depend on the optimization approach discussed in the subsequent sections. Because the RIS controls the spatial direction and strength of the reflected signal, its phase profile plays the role of a passive beamforming vector. In this work, the term "beamforming" therefore refers exclusively to RIS phase control: both the satellites and the user are modeled as single-antenna nodes with fixed radiation patterns and no digital precoding.

At the physical layer, the user receives the downlink signal from each satellite S_m through two principal paths: the LoS path and the two-hop RIS-reflected path. To characterize the direct transmission, the complex channel gain of the LoS link from satellite S_m at time t is expressed as

$$\hat{h}_m^{\mathrm{dir}}(t) = \frac{e^{j2\pi f_c \tau_{\mathrm{su}_m}(t)}}{\sqrt{L_{\mathrm{SU}_m}(t)}}, \tag{7}$$

where $L_{\mathrm{SU}_m}(t)$ and τ_{su_m} denote the path loss and propagation delay of the satellite–user link, respectively, as previously defined.

The RIS-reflected path, assuming identical amplitude reflection by all RIS elements along with individual phase shifts, is expressed as follows:

$$\hat{h}_m^{\mathrm{ris}}(t) = \frac{\sqrt{G_{\mathrm{RIS}}(N)}}{g_{\mathrm{SR}_m}(t) g_{\mathrm{RU}}(t)}\ \frac{1}{N}\left(\sum_{n=1}^{N} e^{j\phi_n(t)}\right), \tag{8}$$

where $g_{\mathrm{SR}_m}(t) = \sqrt{L_{\mathrm{SR}_m}(t)}\, e^{-j2\pi f_c \tau_{\mathrm{SR}_m}(t)}$, $g_{\mathrm{RU}}(t) = \sqrt{L_{\mathrm{RU}}(t)}\, e^{-j2\pi f_c \tau_{\mathrm{RU}}(t)}$, and $G_{\mathrm{RIS}}(N)$ denotes the power-domain array gain provided by the RIS, which is defined as

$$G_{\mathrm{RIS}}(N) = \eta_{\mathrm{RIS}}\, N^2\, L_q(b). \tag{9}$$

The parameter $\eta_{\mathrm{RIS}} \in (0, 1]$ captures non-ideal hardware effects such as ohmic and dielectric losses, while $L_q(b) = \mathrm{sinc}^2\left(\frac{\pi}{2^b}\right)$, describes the deterministic attenuation introduced by $b-$bit phase over $[0, 2\pi)$ [3]. Practical implementations, therefore, experience two principal non-idealities: element efficiency loss, represented by η_{RIS}, and quantization loss, embedded in $L_q(b)$ [21].

In the ideal limit of loss-free hardware with continuous phase tuning, $\eta_{\mathrm{RIS}} = L_q(b) = 1$,and full phase alignment $\phi_n(t) = \phi^\star$ for all n yield $\frac{1}{N}\sum_{n=1}^{N} e^{j\phi^\star} = 1$. The reflected channel then reduces to $\hat{h}_m^{\mathrm{ris}}(t) = \frac{N}{g_{\mathrm{SR},m}(t)\, g_{\mathrm{RU}}(t)}$, such that the field amplitude scales linearly with the number of elements, i.e., $|\hat{h}_m^{\mathrm{ris}}(t)| \propto N$ (field gain), and the received power scales quadratically with $|\hat{h}_m^{\mathrm{ris}}(t)|^2 \propto N^2$.

By linear superposition, the effective channel between satellite S_m and the user is

$$\hat{h}_{\mathrm{eff},m}(t) = \hat{h}_m^{\mathrm{dir}}(t) + \hat{h}_m^{\mathrm{ris}}(t) \tag{10}$$

Given the physical configuration and propagation models above, in the next section, we will describe how the user's mobility is predicted and incorporated into system optimization.

Modelling assumptions and scope.: The above propagation model focuses on the dominant geometry- and RIS-related effects in a single-beam (or orthogonally scheduled multi-beam) LEO downlink. We assume that any residual inter-beam interference and satellite payload distortions can be absorbed into the effective noise term in the SINR expression, and that the satellite transmit chain operates in its linear region. RIS hardware non-idealities are captured through the

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element efficiency η_{RIS} and the phase-quantization loss $L_q(b)$ in $G_{\mathrm{RIS}}(N)$. We also adopt a standard block-fading model: the composite channels $h_{\mathrm{eff},m}(t)$ are quasi-static within each decision interval of duration Δt and are updated from slot to slot according to the mobility and prediction model in Section III. For the LEO geometry and user speeds considered in this work, Δt is chosen significantly smaller than the channel coherence time, so this approximation is well justified.

III. User Mobility and State Space Representation

A core challenge in real-time RIS phase control for dynamic LEO satellite networks is anticipating future user positions and the corresponding satellite-to-user channel conditions. This predictive capability is essential for planning RIS actions that are optimal not only for the present moment but also for a future prediction horizon denoted by $t+\hbar$. Here, $\hbar$ specifies the look-ahead interval—namely, the period in the future over which user mobility and channel states are forecasted. This section develops the mathematical framework for modeling user mobility, estimating users' future locations, and predicting the corresponding channel coefficients for each satellite. These predicted quantities serve as critical inputs to the intelligent RIS control strategy detailed in Section V-C.

A. User Mobility Model

The horizontal motion of the mobile user U is described on a local tangent plane using a state vector that captures both position and velocity:

$$\mathbf{x}(t) = \begin{bmatrix} x(t) & \dot{x}(t) & y(t) & \dot{y}(t) \end{bmatrix}^{\mathrm{T}} \in \mathbb{R}^4, \tag{11}$$

where $x(t)$ and $y(t)$ denote the user's planar coordinates, while $\dot{x}(t)$ and $\dot{y}(t)$ represent the corresponding velocity components. The user's state then evolves according to the following discrete-time linear dynamic system:

$$\mathbf{x}(t+\Delta t) = \underbrace{\mathbf{F}}_{\substack{\text{motion}\\ \text{model}}} \mathbf{x}(t) + \underbrace{\mathbf{G}}_{\substack{\text{process}\\ \text{noise shaping}}} \mathbf{w}(t) \tag{12}$$

where $\mathbf{w}(t) \sim \mathcal{N}(0, q\mathbf{I}_2)$ is a zero-mean Gaussian process noise vector that models random user accelerations and other minor, unpredictable variations in motion. The scalar parameter q regulates the intensity of the noise.

The user state transition matrix ($\mathbf{F}$) and the noise shaping matrix ($\mathbf{G}$) are defined as follows:

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{G} = \frac{\Delta t^2}{2} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}. \tag{13}$$

This model is based on the widely used constant-velocity assumption, which is well justified for mobility prediction over short time intervals and moderate user speeds [22]. Deviations from ideal constant-velocity motion, such as user acceleration or minor trajectory changes, are captured by the process noise term $\mathbf{w}(t)$. This enables systematic and robust predictions of future user positions for real-time RIS control.

B. State Estimation and Prediction via Extended Kalman Filter

To enhance mobility prediction and reduce state estimation uncertainty, the user's state is recursively inferred from direction-of-arrival (DoA) and time-of-flight (ToF) measurements, which are extracted from pilot signals transmitted by multiple satellites. In this framework, it is assumed that the user terminal is equipped with suitable spatial processing and high-precision timing capabilities, enabling it to locally estimate the corresponding azimuth (α_m), elevation (β_m), and propagation delay (τ_m) associated with the received pilot signal for each visible satellite (S_m).

At each time step, measurements from all M accessible satellites are consolidated into a composite measurement vector.

$$\mathbf{z}(t) = \begin{bmatrix} \alpha_1(t) & \beta_1(t) & \tau_1(t)... & \alpha_M(t) & \beta_M(t) & \tau_M(t) \end{bmatrix}^\top \in \mathbb{R}^{3\mathbb{M}}. \tag{14}$$

The acquisition of these measurements is facilitated by the typical geometry of LEO mega-constellations, where multiple satellites are generally within the user's field of view.

Since the measurement process is nonlinear with respect to the user's state, an Extended Kalman Filter (EKF) is employed to recursively estimate the state. The EKF operates in two principal steps:

- Prediction: The state and covariance are propagated forward in time according to the mobility model.
- Update: New satellite observations are incorporated using the linearized measurement model (Jacobian matrix $\mathbf{H}(t)$) to refine the state estimate and error covariance.

Letting $\mathbf{x}(t)$ be the a posteriori state estimate and $\mathbf{P}(t)$ its covariance, the EKF updates are

- Prediction Step:

$$\hat{\boldsymbol{x}}^-(t+\Delta t) = \mathbf{F}\hat{\mathbf{x}}(t), \quad \mathbf{P}^-(t+\Delta t) = \mathbf{F}\mathbf{P}(t)\,\mathbf{F}^\top + q\,\mathbf{G}\,\mathbf{G}^\top. \tag{15}$$

- Update (on receiving new measurement $\mathbf{z}(t+\Delta t)$

$$\begin{gathered} \mathbf{H}(t) = \left.\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}^-(t+\Delta t)} \\ \mathbf{K}(t+\Delta t) = \mathbf{P}^-(t+\Delta t)\mathbf{H}^\top \left[\mathbf{H}\mathbf{P}^-\mathbf{H}^\top + \mathbf{R}\right]^{-1}, \\ \hat{\mathbf{x}}(t+\Delta t) = \hat{\mathbf{x}}^-(t+\Delta t) + \mathbf{K}(t+\Delta t)\left[\mathbf{z}(t+\Delta t) - \mathbf{h}\left(\hat{\mathbf{x}}^-(t+\Delta t)\right)\right], \\ \mathbf{P}(t+\Delta t) = \left[\mathbf{I} - \mathbf{K}(t+\Delta t)\mathbf{H}(t)\right]\mathbf{P}^-(t+\Delta t), \end{gathered} \tag{16}$$

where $\mathbf{K}$ is the EKF gain.

The EKF continuously provides updated estimates of the user's position and velocity.

C. Prediction of Future User State

Given the estimated state at time t, the user's predicted future position at time $t+\hbar$ is obtained by recursively applying the state model:

$$\hat{\mathbf{x}}(t+\hbar) = \mathbf{F}^{\hbar/\Delta t}\,\hat{\mathbf{x}}(t), \qquad , \tag{17}$$

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Then, the predicted 2D position becomes as follows:

$$\hat{\mathbf{p}}_{\hbar} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \hat{\mathbf{x}}(t+\hbar) \tag{18}$$

where $\hat{\mathbf{p}}_{\hbar} \in \mathbb{R}^2$ is the predicted user position at time $t+\hbar$.

D. Channel Prediction at the Planning Horizon

Given the predicted user position $\hat{\mathbf{p}}_{\hbar}$, all relevant distances, delays, and channel coefficients can be evaluated at the planning horizon $t+\hbar$. This enables the computation of the expected channel gains for each satellite.

The predicted direct channel coefficient from satellite S_m to the user is expressed as:

$$\hat{h}_{\mathrm{m}}^{\mathrm{dir}}(t+\hbar) = \frac{e^{j2\pi f_c \tau_{\mathrm{SU}_m}(t+\hbar)}}{\sqrt{L_{\mathrm{SU}_m}(t+\hbar)}}, \tag{19}$$

where $\tau_{\mathrm{SU}_m}(t+\hbar)$ and $L_{\mathrm{SU}_m}(t+\hbar)$ denote the propagation delay and path loss, respectively, both evaluated at the predicted user position at time $t+\hbar$.

Similarly, the case involving RIS assistance is represented as follows:

$$\hat{h}_{\mathrm{m}}^{\mathrm{ris}}(t+\hbar) = \frac{\sqrt{G_{\mathrm{RIS}}(N)}}{g_{\mathrm{SR}_m}(t+\hbar)\, g_{\mathrm{RU}}(t+\hbar)} \frac{1}{N} \left(\sum_{n=1}^{N} e^{j\phi_n(\hbar)} \right), \tag{20}$$

where $g_{\mathrm{SR}_m}(t+\hbar)$ and $g_{\mathrm{RU}}(t+\hbar)$ represent the channel gains from the satellite S_m to the RIS and from the RIS to the user, respectively, both computed at the planning horizon. Additionally, $\phi_n(\hbar)$ is the planned phase shift for the RIS element n at time $t+\hbar$.

The total predicted effective channel from satellite S_m to the user at the planning horizon is thus given by:

$$\hat{h}_{\mathrm{eff,\,m}}(t+\hbar) = \hat{h}_{\mathrm{m}}^{\mathrm{dir}}(t+\hbar) + \hat{h}_{\mathrm{m}}^{\mathrm{ris}}(t+\hbar). \tag{21}$$

The vector $\hat{\mathbf{h}}_{\mathrm{eff}}(t+\hbar) \triangleq \left[\hat{h}_{\mathrm{eff,\,1}}(t+\hbar), ..., \hat{h}_{\mathrm{eff,\,M}}(t+\hbar)\right]^{\mathrm{T}}$ serves as the input for the RIS control and optimization procedures in the following sections. On the basis of these predictions, Section IV formulates a Markov Decision Process (MDP) and trains a TD3 agent for real-time phase optimization.

To evaluate the accuracy of the composite channel prediction, we employ the normalized mean square error (NMSE), which is defined as:

$$\mathrm{NMSE} = \frac{\mathbb{E}\left[\left\|\hat{\mathbf{h}}_{\mathrm{eff}}(t+\hbar) - \mathbf{h}_{\mathrm{eff}}(t+\hbar)\right\|_2^2\right]}{\mathbb{E}\left[\left\|\mathbf{h}_{\mathrm{eff}}(t+\hbar)\right\|_2^2\right]}, \tag{22}$$

where $\hat{\mathbf{h}}_{\mathrm{eff}}(t+\hbar)$ is the vector of predicted channel coefficients from all M satellites, and $\mathbf{h}_{\mathrm{eff}}(t+\hbar)$ is the corresponding true channel vector. The expectation is taken over channel realizations and prediction noise, reflecting the average prediction error relative to channel power. Here, $\hat{\mathbf{h}}_{\mathrm{eff}}(t+\hbar)$ denotes the *predicted and calibrated* composite channel obtained from the EKF-based mobility tracking and the SNR calibration step, rather than perfect instantaneous CSI. Consequently, CSI uncertainty is captured by the NMSE in (22), and the TD3 agent always operates using these imperfect channel predictions.

IV. Learning-Based Real-Time RIS Phase Optimization

In RIS-assisted dynamic LEO networks, multiple performance objectives can be considered, including throughput, latency, and reliability. In this work, we focus on *energy efficiency (EE)* due to the stringent power constraints of LEO platforms and the non-negligible energy cost of satellite transmission and RIS operation. Optimizing a physically grounded EE metric enables improved link performance while limiting power consumption, thereby supporting scalable multi-satellite operation in 6G non-terrestrial networks.

Given the predicted user mobility and channel evolution, the RIS phase configuration must be updated in real time. Since geometry and channels vary continuously, the control problem is inherently sequential: at each time step, a new phase vector is selected to maximize EE under time-varying conditions. Static or one-shot optimizers are therefore inadequate. We instead formulate RIS phase control as a Markov decision process (MDP) and solve it using deep reinforcement learning (DRL), allowing adaptive phase updates based on real-time feedback and uncertainty. The MDP components—state $\mathbf{s}_t$, action $\mathbf{a}_t$, and transition dynamics—are defined below.

The resulting controller operates in a closed loop: user motion is predicted, channels are estimated, the state is constructed, the RIS phase vector is selected via the learned policy, and new measurements are incorporated for the next step, enabling robust and energy-efficient operation in dynamic LEO environments.

Formally, we cast the RIS phase-control problem as a continuous-state, continuous-action Markov decision process (MDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$. At each decision instant t, the system occupies a state $\mathbf{s}_t \in \mathcal{S}$, the RIS controller selects an action $\mathbf{a}_t \in \mathcal{A}$ (the next phase vector), the environment evolves according to the transition kernel $P(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t)$, and the controller receives a scalar reward $r_t = r(\mathbf{s}_t, \mathbf{a}_t)$ that reflects the instantaneous energy efficiency. The precise definitions of state $\mathbf{s}_t$, action $\mathbf{a}_t$, and reward r_t used in this work are given in the subsections below, see in particular (23) for the state, (25) for the action space, and (34)–(35) for the reward. To solve this MDP we adopt the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm, an off-policy actor–critic method designed for continuous control. The actor network π_ψ implements a deterministic policy $\mathbf{a}_t = \pi_\psi(\mathbf{s}_t)$, while the critic networks approximate the action-value function $Q(\mathbf{s}_t, \mathbf{a}_t)$ and are trained with the TD3 twin-critic and target-network updates described in Section V-B and Section V-C.

A. State Definition

At each decision instant t, the agent corresponding to the RIS controller is responsible for selecting the next phase configuration of the RIS. To make an informed decision, the agent observes the current system state, denoted by $\mathbf{s}_t$. This state vector consolidates all relevant real-time information accessible to the controller, including the predicted user position at the future planning horizon ($\hat{\mathbf{p}_{\hbar}}$), the predicted composite channel gains from each satellite to the user ($\hat{\mathbf{h}}_{\mathrm{eff}}(t+\hbar)$), and the current RIS phase configuration ($\boldsymbol{\phi}(t)$). Mathematically, the system state is represented as:

$$\mathbf{s}_t = \Big(\underbrace{\hat{\mathbf{p}_{\hbar}}}_{\substack{\text{predicted}\\ \text{2D user position}}}, \underbrace{\hat{\mathbf{h}_{\text{eff}}}(t+\hbar)}_{\substack{\text{predicted}\\ \text{channels}}}, \underbrace{\boldsymbol{\phi}(t)}_{\substack{\text{current}\\ \text{RIS phase}}} \Big) \in \mathcal{S}, \tag{23}$$

where $\hat{\mathbf{p}_{\hbar}} \in \mathbb{R}^2$ is the 2D user position predicted via the EKF, $\hat{\mathbf{h}_{\text{eff}}}(t+\hbar) \in \mathbb{C}^M$ is a vector containing the predicted complex channel gains from all satellites, and $\boldsymbol{\phi}(t)$ is the current RIS phase vector: $\boldsymbol{\phi}(t) = [\phi_1(t), \phi_2(t), \ldots, \phi_N(t)]^{\mathrm{T}} \in [0, 2\pi)^N$.

B. Action Space

Given the observed system state, the RIS controller (agent) selects an action $\mathbf{a}_t$ that specifies the RIS phase configuration to be implemented at the next planning horizon. Specifically, the action is defined as a vector of phase shifts to be applied to N RIS elements:

$$\mathbf{a}_t = \boldsymbol{\phi}(t+\hbar) = \begin{bmatrix} \phi_1(t+\hbar) \\ \phi_2(t+\hbar) \\ \vdots \\ \phi_N(t+\hbar) \end{bmatrix} \in \mathcal{A}. \tag{24}$$

Each entry $\phi_n(t+\hbar)$ in this vector determines the reflection phase imposed by the n^{th} RIS element. These phases control how incoming signals are redirected by the RIS and, thus, play a crucial role in enhancing the desired signal for the user.

The action space $\mathcal{A}$ defines the set of valid phase vectors from which the agent can choose. This space depends on the hardware capabilities of the RIS:

• In the quantized (discrete) case, each RIS element can impose only a limited number of phase values. If the resolution is L levels, then each ϕ_n must be selected from the set $\phi_n(t+\hbar) \in \left\{0,\ \frac{2\pi}{L}, \ldots, \frac{2\pi(L-1)}{L}\right\}$. In this case, the action space is:

$$\mathcal{A} = \left\{ [0, 2\pi)^N \,\middle|\, \phi_n(t+\hbar) \in \left\{0,\ \frac{2\pi}{L}, \ldots, \frac{2\pi(L-1)}{L}\right\}, \right. \\ \left. \forall_n \in \{1, \ldots, N\} \right\}. \tag{25}$$

• In the continuous case, each RIS element can adjust its phase to any real value in the interval $[0, 2\pi)$, so that the action $\mathbf{a}_t$ fully determines the RIS phase profile applied at time $t+\hbar$. This results in a continuous action space given by $\mathcal{A} = [0, 2\pi)^N$.

Thus, the action $\mathbf{a}_t$ fully encodes the phase configuration that the RIS will apply at the prediction horizon. This choice directly affects how reflected signals from multiple satellites combine at the user terminal, either constructively or destructively, making it a central factor in controlling both signal quality and energy efficiency. Selecting the optimal phase configuration at each step is, therefore, a critical task for the RIS controller, especially in highly dynamic satellite communication environments.

C. Transition Dynamics

The evolution of the system state from s_t to s_{t+1} is governed by the sequential processes of user mobility estimation, channel prediction, and RIS phase assignment. At each time step, the user's updated state vector $\mathbf{x}(\mathbf{t}+\boldsymbol{\Delta}\mathbf{t})$ is obtained via the EKF recursion, as defined in Equation 15. This state is then propagated by the planning horizon $\hbar$ to yield the predicted user position $\hat{\mathbf{p}_{\hbar}}(t+\Delta t) = \mathbf{P}\,\mathbf{F}^{\hbar/\Delta \mathrm{t}}\hat{\mathbf{x}}(t+\Delta t)$. Based on this predicted position, the effective channel coefficients for all satellites, $\hat{\mathbf{h}_{\text{eff}}}(t+\Delta t+\hbar)$, are recalculated deterministically according to the models established in Section III-D, incorporating both direct and RIS-assisted paths with the current RIS configuration. The RIS phases for the next time step are then set as $\phi(t+\Delta t) = \mathrm{a}_t$, where $\mathbf{a}_t$ is the action chosen by the agent. Collectively, the new state is assembled as follows:

$$\mathbf{s}_{t+1} = \Big(\hat{\mathbf{p}_{\hbar}}(t+\hbar), \hat{\mathbf{h}_{\text{eff}}}(t+\Delta t+\hbar), \mathbf{a}_t \Big), \tag{26}$$

with the transition kernel $P(\mathrm{s}_{t+1} \mid \mathrm{s}_t, \mathrm{a}_t)$ encapsulating the stochastic nature of the EKF update and the deterministic mappings of channel prediction and RIS phase assignments. Formally, the overall state transition law is expressed as

$$\begin{aligned} P\big(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t\big) = &\underbrace{P_{\text{EKF}}\big(\hat{\mathbf{x}}(t+\Delta t) \mid \hat{\mathbf{x}}(t)\big)}_{\text{EKF prediction (Gaussian)}} \\ &\underbrace{\delta\left(\hat{\mathbf{p}}_{\hbar}(t+\Delta t) - \mathbf{F}^{\hbar/\Delta t}\hat{\mathbf{x}}(t+\Delta t)\right)}_{\text{Position propagation}} \\ &\underbrace{\delta\left(\hat{\mathbf{h}}_{\text{eff}}(t+\Delta t+\hbar) - h(\hat{\mathbf{p}}_{\hbar}(t+\Delta t), \mathbf{a}_t)\right)}_{\text{Channel prediction}} \\ &\underbrace{\delta\big(\boldsymbol{\phi}(t+\Delta t) - \mathbf{a}_t\big)}_{\text{RIS-phase assignment}} \end{aligned} \tag{27}$$

where P_{EKF} denotes the Gaussian prediction density arising from the process noise in the user mobility model, and each Dirac delta function δ, enforces the deterministic mappings inherent in the prediction and control architecture. This structure enables the agent to adaptively track user motion and optimally control the RIS configuration in real time.

After defining the state, action, and transition dynamics of the system, we now address the analytical and learning-based optimization of energy efficiency. The entire closed-loop procedure, including user mobility prediction, channel estimation, and real-time RIS phase control with deep reinforcement learning,is systematically presented in Algorithm 1. This algorithm outlines each step, from environmental sensing and EKF-based forecasting to the selection and application of RIS phase shifts by the DRL agent. In the following section, we introduce the energy efficiency metric and explain how it is incorporated into the reinforcement learning framework for real-time optimization.

V. Energy-Efficiency Analysis and Optimization

Energy efficiency has emerged as a critical performance criterion in RIS-assisted satellite communication networks, directly influencing both sustainability and operational costs.

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Algorithm 1: Closed-Loop DRL-Based RIS Phase-Control Process

Input: Initial state $s_0 = (\hat{p}_\hbar, \hat{h}_{\text{eff}}, \boldsymbol{\phi})$, Trained DRL policy π_ψ, Environment parameters
Output: Real-time RIS phase configuration $\boldsymbol{\phi}(t)$
for *each time step* $t = 0, 1, 2, \ldots$ **do**
  Obtain channel and satellite measurements (DoA, ToF)
  `// Predict user mobility using EKF`
  Update user state estimate $\hat{\mathbf{x}}(t)$ and covariance $P(t)$ via EKF
  Predict user position at planning horizon: $\hat{p}_\hbar(t)$
  Predict composite channel $\hat{h}_{\text{eff}}(t+\hbar)$ based on predicted position and channel model
  Construct current system state $s_t = (\hat{p}_\hbar(t), \hat{h}_{\text{eff}}(t+\hbar), \boldsymbol{\phi}(t))$
  `// RIS Phase Decision`
  Compute next RIS phase configuration: $\boldsymbol{\phi}(t+\hbar) \leftarrow \pi_\psi(s_t)$
  Apply $\boldsymbol{\phi}(t+\hbar)$ to RIS hardware
  Receive reward (energy efficiency) r_t and observe new measurements
  `// Feedback`
  Update replay buffer and, if training, update actor and critic networks

In such systems, EE is defined as the ratio of the delivered information rate to the total consumed power, which includes contributions from satellite transmitters, RIS hardware, and supporting signal processing infrastructure. Unlike conventional fixed-parameter optimization, the dynamic and stochastic nature of LEO satellite networks, compounded by user mobility and rapidly varying channels, necessitates an intelligent, adaptive approach. In this work, we address this challenge by embedding the analytical EE metric into a DRL framework, thereby enabling the autonomous and real-time optimization of RIS phase configurations under non-stationary conditions.

The following subsections first formulate the EE metric in terms of physical and system parameters, then describe the design of the DRL reward and learning architecture, and finally establish the link between theoretical and data-driven optimization.

A. Energy-Efficiency Metric as Reinforcement Learning Reward

In the context of a RIS-assisted satellite downlink system, EE is defined as the number of information bits successfully delivered per joule of total power consumed across all system components. This metric incorporates not only the transmit power radiated by each satellite but also the circuit power associated with active RIS elements and the fixed signal processing overhead. At the planning horizon $t+\hbar$, the RIS controller can therefore be viewed as solving an instantaneous EE maximization over the per-satellite transmit powers $\{P_m(t+\hbar)\}$ and the RIS phase vector $\boldsymbol{\phi}(t+\hbar)$. For a given RIS configuration and predicted composite channels $\hat{h}_{\text{eff},m}(N, t+\hbar)$, the SINR of satellite S_m is

$$\gamma_m(N, t+\hbar) = \frac{P_m(t+\hbar)\left|\hat{h}_{\text{eff},m}(N, t+\hbar)\right|^2}{\sigma^2},$$

and the target rate R_{req} is guaranteed by enforcing $\gamma_m(N, t+\hbar) \geq \gamma_{\text{req}}$ for all $m = 1, \ldots, M$, where $\gamma_{\text{req}} = 2^{R_{\text{req}}/B} - 1$ is the required SINR. In this setting, the denominator of the EE metric consists of the sum of the per-satellite transmit powers, the RIS circuit power, and the fixed processing overhead, and the underlying optimization can be interpreted as maximizing R_{req} divided by this total power subject to the SINR constraints and the feasible set of RIS phases. For the single-user cooperative downlink considered here, and for fixed predicted channels and RIS phases, these SINR constraints are active at the optimum, which leads directly to the minimum required powers in (29). Substituting $P_m^{\min}(N, t+\hbar)$ into the total power expression yields the following compact closed-form EE expression.

At the planning horizon $t+\hbar$, the instantaneous EE is given by

$$\mathrm{EE}(N, t+\hbar) = \frac{B\,\log_2(1+\gamma_{\text{req}})}{\displaystyle\sum_{m=1}^{M} \frac{\gamma_{\text{req}}\,\sigma^2}{\left|\hat{h}_{\text{eff},m}(N, t+\hbar)\right|^2} + N_{\text{act}}(t,\hbar)\,P_{\text{el}} + P_{\text{proc}}}, \tag{28}$$

where B denotes the channel bandwidth, $\gamma_{\text{req}} = 2^{R_{\text{req}}/B} - 1$ is the minimum required SINR for a given per-user rate R_{req}, and σ^2 is the user-side noise power. The effective composite channel gain from satellite m, denoted as $\hat{h}_{\text{eff},m}(N, t+\hbar)$, is predicted at the planning horizon based on the current RIS phase configuration, the most recent EKF-based mobility prediction, and the associated channel models, as described in Section III. The terms $N_{\text{act}}(t,\hbar)$ and P_{el} denote the number of active RIS elements and the per-element circuit power, respectively, while P_{proc} represents the baseband processing overhead.

The minimum transmission power required by each satellite to satisfy the user's SINR requirement is expressed as

$$P_m^{\min}(N, t+\hbar) = \frac{\gamma_{\text{req}}\,\sigma^2}{\left|\hat{h}_{\text{eff},m}(N, t+\hbar)\right|^2}, \tag{29}$$

and the total radiated power is $P_{\text{tot}} = \sum_{m=1}^{M} P_m^{\min}(N, t+\hbar)$.

The aggregate RIS circuit power is determined adaptively as follows:

$$P_{RIS}(t+\hbar) = N_{\text{act}}(t,\hbar)\,P_{\text{el}}, \tag{30}$$

where the activity indicator for element n is defined as

$$u_n(t,\hbar) = \begin{cases} 1, & \max\limits_m \left|\dfrac{\partial \hat{h}_{\text{eff},m}(N, t+\hbar)}{\partial \phi_n}\right| > \xi, \\ 0, & \text{otherwise}, \end{cases} \tag{31}$$

and $N_{\text{act}}(t,\hbar) = \sum_{n=1}^{N} u_n(t,\hbar)$, for a design threshold ξ.

Remark 1 (Threshold ξ): The threshold ξ governs the energy efficiency–power trade-off by controlling the number of active RIS elements N_{act}, and hence the circuit power $P_{\text{RIS}} =$

$N_{\text{act}}P_{\text{el}}$. A larger ξ activates fewer elements (reducing circuit power), while a smaller ξ activates more elements (increasing beamforming gain at the expense of power consumption). A practical selection rule is to set ξ as the $(1-\rho)$-quantile of the set $\{\max_m |\partial\hat{h}_{\text{eff},m}/\partial\phi_n|\}_{n=1}^N$, which yields $N_{\text{act}} \approx \rho N$. We further perform a sensitivity analysis by sweeping ξ (equivalently ρ) and evaluating EE versus N_{act}/N, which reveals a broad optimum region under realistic per-element circuit power constraints.

If the channel gains from each satellite to the user are approximately equal and symmetric power allocation is adopted, the minimum required transmit power per satellite can be approximated as

$$P_m^{\min}\big(N, t+\hbar\big) \approx \frac{1}{M}\sum_{i=1}^{M} P_i^{\min}\big(N, t+\hbar\big). \tag{32}$$

This approximation is most accurate when the satellites are similarly positioned with respect to the RIS and the user, such as in dense constellations or when the user is centrally located relative to the satellite cluster. Although channel disparities are inevitable in practical systems, this assumption simplifies the analysis and enables the benchmarking of idealized performance. In real deployments, however, power allocation can be further refined to account for differences in path loss among the satellites.

As the number M of cooperating satellites increases, the required transmit power for each individual satellite $P_m^{\min}$ decreases while maintaining a fixed total power requirement. In this context, power allocation refers to the process by which the Network Operations Center (NOC) assigns transmit power across satellites to collectively satisfy the user's SINR constraints [12].

To provide a meaningful baseline for evaluating the performance of the reinforcement learning (RL) agent, we also consider a *random-phase* RIS benchmark in which the element phase shifts are chosen independently and uniformly at random, i.e., $\phi_n \sim \mathcal{U}[0, 2\pi)$. Writing $\hat{h}_{\text{eff},m} = \hat{h}_{\text{dir},m} + \hat{h}_{\text{ris},m}$ and utilizing the independence of phases, the cross terms vanish in expectation, and the normalized phasor average adheres to $\mathbb{E}\Big[\big|\frac{1}{N}\sum_{n=1}^{N} e^{j\phi_n}\big|^2\Big] = \frac{1}{N}$. Consequently, the expected composite channel *power* for satellite m is

$$\begin{aligned}\mathbb{E}\Big[\big|\hat{h}_{\text{eff},m}(N, t+\hbar)\big|^2\Big] &= L_{SU,m}(t+\hbar) + \\ &\eta_{\text{RIS}}\, L_q(b)\, N\, L_{SR,m}(t)\, L_{RU}(t+\hbar).\end{aligned} \tag{33}$$

Thus, under random phases, there is no coherent N^2 combination; the RIS contribution grows only linearly with N. This random-phase scenario captures the absence of coordinated RIS control and serves as a reference for quantifying the performance gains achieved by the RL-optimized phases.

Within the DRL framework, the instantaneous EE metric serves as the reward for each control action, as will be discussed in detail in Section V-B. Following each RIS phase update, the agent predicts user positions and channel conditions at the planning horizon $t+\hbar$, calculates the corresponding energy efficiency, and receives this value as a scalar reward. By maximizing the expected cumulative reward over time, the agent learns a phase-control policy that balances the nonlinear benefits of RIS scaling with the linear circuit power cost, thus driving the system toward the global EE optimum established by theoretical analysis. This seamless integration of the analytical EE expression into the RL reward structure ensures that the learned RIS control policy directly addresses the system's fundamental power-throughput trade-off.

B. DRL-Based EE Optimization

The real-time optimization of EE in RIS-assisted satellite systems requires a control method that can continuously adapt to rapid fluctuations in user mobility, channel states, and RIS configurations. To address this challenge, we formalize the phase control problem as a Markov Decision Process (MDP), in which the RIS controller acts as an autonomous agent, learning to maximize long-term EE by sequentially interacting with its environment.

Within this MDP, as defined in Section IV, the agent observes the current system state $\mathbf{s}_t$, which comprises the predicted user location, channel gains, and the current RIS phase configuration. It then selects an action $\mathbf{a}_t$ that encodes the RIS phase shifts to be applied at the next planning horizon. The environment responds by transitioning to a new state $\mathbf{s}_{t+1}$, reflecting updated mobility, channel realizations, and RIS settings. It returns a scalar reward r_t corresponding to the system's achieved EE.

1) Reward Function for EE: The instantaneous reward at each time step is defined to directly reflect EE, as follows:

$$r_t^{\text{EE}} = \text{EE}\big(N, t+\hbar\big) - \lambda\,\big\|\boldsymbol{\phi}(t+\hbar) - \boldsymbol{\phi}(t)\big\|_2^2, \tag{34}$$

where $\text{EE}\big(N, t+\hbar\big)$ is the EE at the planning horizon, as defined in (28), and the regularization term penalizes abrupt RIS reconfiguration, with $\lambda > 0$ controlling the trade-off between performance and phase stability.

To also capture alternative design objectives such as throughput or coverage, we introduce a parametric reward family of the form

$$r_t^{(\mu)} = \text{EE}\big(N, t+\hbar\big) + \mu\,\widetilde{R}_t - \lambda\,\big\|\boldsymbol{\phi}(t+\hbar) - \boldsymbol{\phi}(t)\big\|_2^2, \tag{35}$$

where $\widetilde{R}_t = B^{-1}R_t/R_{\text{req}}$ is a normalized throughput-related term constructed from the instantaneous rate $R_t = B\log_2(1+\gamma_t)$ at the prediction horizon, and $\mu \geq 0$ is a weight that balances energy efficiency against the rate/coverage preference. The special case $\mu = 0$ recovers the pure EE-oriented reward in (34), while increasing μ shifts the objective towards higher SNR/throughput, potentially at the cost of additional power consumption.

For each fixed weight μ, the TD3 agent learns a policy $\pi^{(\mu)}$ that maximizes the discounted return associated with $r_t^{(\mu)}$. Under this policy, we evaluate three performance metrics: (i) the long-term average energy efficiency $\text{EE}^{(\mu)}$ (bit J^{-1}), obtained as the time average of $\text{EE}(N, t+\hbar)$ in (28); (ii) the average user throughput $R^{(\mu)} = \mathbb{E}[R_t^{(\mu)}]$; and (iii) the coverage probability $\mathbb{P}\big[\gamma_t^{(\mu)} \geq \gamma_{\text{req}}\big]$, where γ_{req} is the target SINR associated with R_{req}.

Multi-satellite phase coordination: The RIS controller outputs a single phase vector $\boldsymbol{\phi}(t+\hbar)$ that is applied to the same physical surface for all incident satellite signals. Thus, coordination across M satellites is achieved implicitly through centralized control: the state in (23) contains the predicted multi-satellite channel vector $\hat{\mathbf{h}}_{\text{eff}}(t+\hbar)$, while the EE reward in (28) aggregates the required transmit powers of all satellites. When satellite geometries differ, the learned policy selects an EE-optimal compromise phase profile rather than conflicting per-satellite configurations. Robustness to rapid geometry changes is further improved by EKF look-ahead prediction and by the phase-smoothness penalty in (34), which prevents abrupt reconfiguration.

2) Bellman Equation and DRL Architecture: In this subsection, we instantiate the TD3 actor–critic architecture for the RIS control MDP introduced above and specify the Bellman equation and network update rules used during training. The central goal of the training process is for the RIS controller to learn an optimal phase-control policy, denoted by $\pi_{\boldsymbol{\psi}}$, that maximizes the expected long-term discounted sum of energy–efficiency-based rewards. Here, $\boldsymbol{\psi}$ is the vector of trainable parameters (weights and biases) within the actor neural network, and the performance objective is written as

$$J(\pi_{\boldsymbol{\psi}}) = \mathbb{E}_{\pi_{\boldsymbol{\psi}}}\left[\sum_{\tau=t}^{\infty} \gamma^{\tau-t} r_\tau\right], \tag{36}$$

where $0 < \gamma < 1$ is the discount factor that balances immediate and future performance, and r_τ denotes the instantaneous reward at time τ (e.g., the EE-oriented or parametric reward $r_\tau^{(\mu)}$ in (35)).

The sequential nature of this optimization can be expressed in terms of the action–value function (Q-function)

$$Q^\pi(\mathbf{s}_t, \mathbf{a}_t) = \mathbb{E}\left[\sum_{\tau=t}^{\infty} \gamma^{\tau-t} r_\tau \,\middle|\, \mathbf{s}_t, \mathbf{a}_t\right], \tag{37}$$

which satisfies the Bellman equation

$$Q^\pi(\mathbf{s}_t, \mathbf{a}_t) = \mathbb{E}\big[r_t + \gamma\, Q^\pi\big(\mathbf{s}_{t+1}, \pi_{\boldsymbol{\psi}}(\mathbf{s}_{t+1})\big) \,\big|\, \mathbf{s}_t, \mathbf{a}_t\big], \tag{38}$$

where $\mathbf{s}_t \in \mathcal{S}$ collects the system state (predicted user position, predicted/filtered channel features, current RIS phase configuration), $\mathbf{a}_t \in \mathcal{A}$ is the RIS phase-action vector, and the expectation is taken over the next-state distribution induced by the mobility and channel dynamics.

In the proposed controller, this Bellman equation is approximated by a TD3 actor–critic architecture: two critics $Q_{\boldsymbol{\theta}_1}(\mathbf{s}, \mathbf{a})$ and $Q_{\boldsymbol{\theta}_2}(\mathbf{s}, \mathbf{a})$, and a deterministic actor $\mu_{\boldsymbol{\psi}}(\mathbf{s})$ that outputs the RIS phase vector. For a minibatch of transitions $(\mathbf{s}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1}, d_t)$, where d_t indicates episode termination, the TD3 target is

$$y_t = r_t + \gamma(1 - d_t) \min_{i=1,2} Q_{\bar{\boldsymbol{\theta}}_i}\big(\mathbf{s}_{t+1}, \mu_{\bar{\boldsymbol{\psi}}}(\mathbf{s}_{t+1}) + \boldsymbol{\varepsilon}_t\big), \tag{39}$$

where $(\bar{\boldsymbol{\theta}}_i, \bar{\boldsymbol{\psi}})$ are slowly updated target-network parameters and $\boldsymbol{\varepsilon}_t$ is the clipped target-policy noise used by TD3 to improve robustness. Each critic minimizes the squared error $\mathbb{E}[(Q_{\boldsymbol{\theta}_i}(\mathbf{s}_t, \mathbf{a}_t) - y_t)^2]$, while the actor is updated by ascending the gradient of $Q_{\boldsymbol{\theta}_1}(\mathbf{s}, \mu_{\boldsymbol{\psi}}(\mathbf{s}))$. All target networks are updated by standard soft updates. In this way, the TD3 training procedure is explicitly linked to the energy-efficiency metric in (28) through the pure EE reward in (34) and its parametric extension in (35).

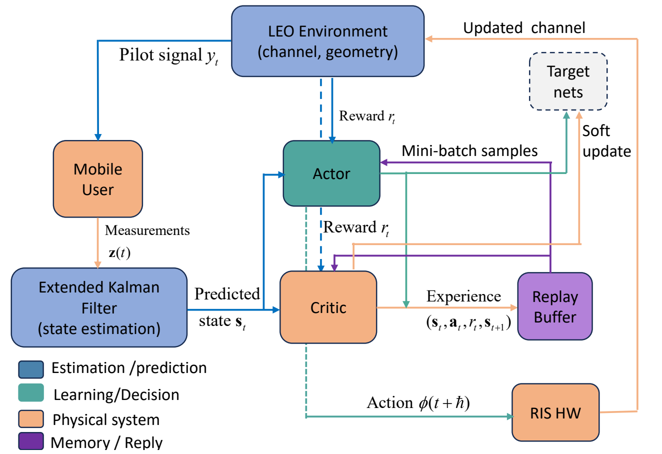

Fig. 2. Block diagram of the EKF- TD3 architecture for real-time RIS phase control in a multi-satellite LEO downlink. The system integrates user mobility prediction via an EKF with adaptive RIS configuration using a TD3-based DRL agent. ”Target nets”refer to the slowly-updated target actor and target critic networks used for stable policy and value updates in the TD3 algorithm (see Section V-C).

In practical RIS-assisted satellite systems, both the state and action spaces are high-dimensional and continuous, making direct dynamic programming intractable. To address this, we adopt a DRL framework, which is illustrated in Fig. 2, where neural networks serve as function approximators for both the policy and value (Q) functions. The DRL architecture maintains two core neural networks: the actor network, parameterized by ψ, which implements the deterministic policy $\boldsymbol{\phi}_t = \pi_\psi(\mathbf{s}_t)$, and the critic network, parameterized by φ, which estimates the action-value function $Q\varphi(\mathbf{s}_t, \boldsymbol{\phi}_t)$ and predicts the expected cumulative discounted reward.

For robust and stable learning, the target networks, namely the target actor π_{ψ^-} and the target critic Q_{φ^-}, are maintained as slowly updated copies of the main networks, providing stable targets for temporal-difference learning. The critic network is trained to minimize the mean squared Bellman error using the following loss function:

$$\mathcal{L}(\varphi) = \mathbb{E}\left[\left(r_t + \gamma\, Q_{\varphi^-}\left(\mathbf{s}_{t+1}, \pi_{\psi^-}(\mathbf{s}_{t+1})\right) - Q_\varphi(\mathbf{s}_t, \mathbf{a}_t)\right)^2\right] \tag{40}$$

Here, the target value $Q_{\varphi^-}\left(\mathbf{s}_{t+1}, \pi_{\psi^-}(\mathbf{s}_{t+1})\right)$ is evaluated using slowly updated target networks, which ensure stable temporal-difference updates.

The actor network is updated by ascending the gradient of the critic's output with respect to its parameters:

$$\psi \leftarrow \psi + \eta \nabla_\psi Q_\varphi(\mathbf{s}, \pi_\psi(\mathbf{s}_t)), \tag{41}$$

where η is the learning rate, a positive scalar that determines the step size for updating the actor's parameters. The term $\nabla_\psi Q_\varphi(s, \pi_\psi(\mathbf{s}_t))$ denotes the gradient of the critic's action-value function with respect to the actor parameters, quantifying the sensitivity of the expected return to changes in the policy.

This update mechanism ensures that the actor network continually improves its policy by directly optimizing for higher EE, as predicted by the critic. Through repeated interactions with the environment and continuous parameter refinement, the agent converges to a phase-control policy that is robust against user mobility, channel fading, and time-varying system dynamics.

By directly embedding the analytic EE metric as the reward signal, the DRL-based RIS controller learns to adaptively select phase configurations that maximize energy efficiency, even under the highly dynamic and uncertain conditions typical of LEO satellite networks.

C. Policy Extraction and RIS Phase Optimization

Building on the DRL formulation in Section V-B, we aim to learn an RIS phase-control policy that attains near-optimal EE under user mobility, time-varying channels, and practical hardware constraints. Since RIS phase control involves a continuous and high-dimensional action space (especially for large arrays), we employ the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm as the learning agent [23]. TD3 is an actor–critic method tailored to continuous control and provides improved training stability and sample efficiency over earlier deterministic policy-gradient schemes [24], making it suitable for real-time RIS phase optimization.

The TD3 training and deployment workflow is summarized in Algorithm 2. The actor network with parameters ψ outputs a deterministic mapping from the observed state to the RIS phase vector (action), while the critic network with parameters φ estimates the expected discounted return of each state–action pair using the EE-based reward in Section V-B1. The critic guides the actor updates by encouraging actions that maximize the predicted long-term EE, yielding an adaptive policy that can be executed online to update RIS phases in real time.

The optimal RIS phase control policy is formally expressed as:

$$\pi_{\psi}^{*}(\mathbf{s}) = \arg\max_{\mathrm{a}\in\mathcal{A}} Q^{*}(\mathbf{s},\mathbf{a}), \tag{42}$$

where $Q^{*}(\mathbf{s},\mathbf{a})$ denotes the optimal action-value function, i.e., the maximum expected sum of discounted energy-efficiency rewards starting from state $\mathbf{s}$, executing action $\mathbf{a}$, and subsequently following the optimal policy.

The Bellman optimality equation for Q^{*} is

$$Q^{*}(\mathbf{s},\mathbf{a}) = r(\mathbf{s},\mathbf{a}) + \gamma\, \mathbb{E}_{\acute{\mathbf{s}}}[\max_{\acute{\mathbf{a}}} Q^{*}(\acute{\mathbf{s}},\acute{\mathbf{a}}) \mid \mathbf{s},\mathbf{a}], \tag{43}$$

where $r(\mathbf{s},\mathbf{a})$ represents the immediate reward (energy efficiency), γ denotes the discount factor, and $\acute{\mathbf{s}}$ indicates the subsequent state.

In practical implementations (e.g., with DDPG/TD3), the actor parameters ψ are updated using gradient ascent:

$$\psi \leftarrow \psi + \eta \nabla_{\psi} Q_{\varphi}(\mathbf{s}, \pi_{\psi}(\mathbf{s})),$$

where ∇_{ψ} is the gradient of the critic with respect to the actor's parameters.

At runtime, the trained RIS controller applies the phase configuration recommended by the learned policy, as outlined in Algorithm 2, and continuously updates its actions as user mobility and channel conditions evolve. This integrated approach, which combines model-driven mobility prediction with data-driven policy optimization, effectively maximizes EE in highly dynamic LEO satellite environments and outperforms static or purely model-based solutions.

Algorithm 2: High-Level TD3 Training Loop for RIS Phase Control

```
Input: Actor π_ψ; critics Q_φ1, Q_φ2; targets
       π_ψ⁻, Q_φ1⁻, Q_φ2⁻;
Replay buffer B (cap. N), batch |M|; discount γ;
learning rates (η_ψ, η_φ);
Soft-update rate τ, policy update interval Δ;
Exploration noise σ_n, target smoothing σ_t, noise clip c;
Time slot Δt, planning horizon ħ
Output: Trained actor policy π_ψ
Initialize environment, EKF, RIS phases, and pre-fill
  replay buffer B with short random rollouts.
for e = 1, 2, ..., E do                          // Episodes
    Reset environment and EKF;
    Build initial state s_0 = ( user prediction at t + ħ,
      predicted channel at t + ħ, RIS phases ).
    for t = 0, Δt, ..., T do                     // Time steps
        // Environment step
        Select action a_t by adding exploration noise to
          actor output and clipping to [0, 2π);
        Apply RIS phases for next slot;
        Update EKF with new measurements, predict
          to horizon;
        Build new state s_{t+1} (future user, channel, and
          phases);
        Calculate reward r_t as energy efficiency at
          horizon minus phase-variation cost;
        // Store experience
        Save (s_t, a_t, r_t, s_{t+1}) to replay buffer;
        s_t ← s_{t+1}
        if |B| ≥ |M| then              // Policy update phase
            Sample random mini-batch M from B;
            Add clipped noise to target policy (target
              smoothing);
            Compute TD3 targets using twin-critic
              minimum;
            for j = 1, 2 do
                Update critic j by minimizing TD error
                  over mini-batch;
            if (t/Δt) mod Δ = 0 then   // Delayed policy update
                Update actor by maximizing critic 1's
                  estimate;
                Soft-update all target networks;
```

VI. Results and Discussions

This section evaluates the proposed EKF–TD3 RIS control in a multi-satellite LEO downlink. Unless otherwise stated, the baseline parameters are $f_c = 19$ GHz, $B = 100$ MHz, $R_{\mathrm{req}} = 50$ Mbps, and user noise figure 3 dB. The RIS power model uses $P_{\mathrm{el}} = 5$ mW per element and a fixed processing overhead

TABLE I
KEY SYSTEM AND TD3 AGENT PARAMETERS USED IN SIMULATIONS

Parameter	Value
Carrier frequency (f_c)	19 GHz
Bandwidth (B)	100 MHz
Target user data rate (R_{req})	50 Mbps
User-side noise figure (NF)	3 dB
Per-element RIS power (P_{el})	5 mW
RIS processing overhead (P_{proc})	1 W
Shadowing loss	25 dB
Path lengths (d_{SU}, d_{SR})	400 km
RIS size (N)	64, 128, 256, 512
Number of satellites (M)	2, 4, 8, 12
User–RIS distance grid (d_{RU})	50–500 m
TD3 Agent Hyperparameters	
Replay buffer size	10^5
Batch size	128
Actor/critic learning rate	$1 \times 10^{-4}/1 \times 10^{-3}$
Discount factor (γ)	0.99
Soft-update rate (τ)	0.005
Exploration noise std (σ_n)	0.1
Target smoothing std (σ_t)	0.2
Noise clip (c)	0.5
Policy update interval (Δ)	2

of 1 W. We investigate RIS sizes $N \in \{64, 128, 256, 512\}$ and satellite multiplicities $M \in \{2, 4, 8, 12\}$.

The per-element RIS circuit power is set to $P_{\text{el}} = 5$ mW, in accordance with the widely adopted RIS power-consumption model in [25]. A shadowing (clutter) loss of 25 dB is incorporated based on the 3GPP NTN channel model for Ka-band suburban/rural environments [26].

For each (M, N) configuration, the EKF predicts user mobility and the resulting channels, and the TD3 agent selects the RIS phase vector to maximize the EE in Section V. All system parameters and TD3 hyperparameters are reported in Table I for reproducibility. Each result is averaged over 10^5 independent Monte Carlo realizations.

We benchmark the proposed EKF–TD3 controller against the following baselines under identical channel/mobility/noise realizations: (i) random RIS phases, $\phi_n \sim \mathcal{U}[0, 2\pi)$; (ii) TD3-only control without EKF prediction (reactive); (iii) EKF prediction with a heuristic deterministic beam-steering rule; and (iv) the full EKF–TD3 framework. We also include the analytical RIS configuration in Section V-A as an upper bound, since it assumes perfect CSI and ideal phase alignment; the EKF–TD3 curves use predicted (imperfect) CSI with nonzero SNR-prediction NMSE and RIS hardware losses, and therefore remain below this bound. All learning-based schemes use the same TD3 architecture and hyperparameters in Table I and are evaluated over the same Monte Carlo scenarios, ensuring a fair comparison in Figs. 3, 4, and 10.

Fig. 3 illustrates the incremental energy-efficiency gain, denoted by ΔEE, achieved through optimal RIS phase control compared to random phase settings, as a function of the number of cooperating satellites, M, for several fixed RIS sizes, N, ($N = 64, 128, 256, 512$). Here, ΔEE is computed as the difference in energy efficiency when the RIS phases are optimized by the TD3-based controller compared to when

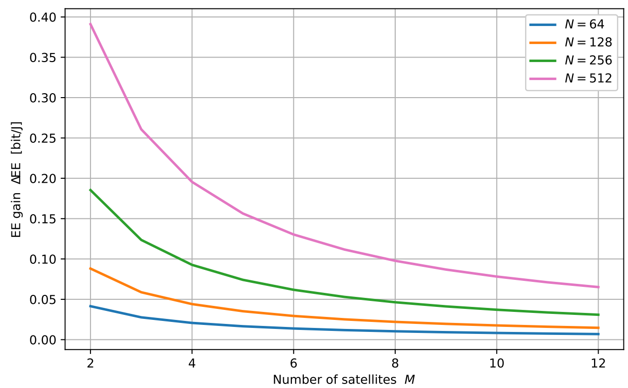

Fig. 3. Energy-efficiency gain ΔEE as a function of the number of satellites M for various RIS element counts N.

they are assigned randomly, i.e.,

$$\Delta \text{EE} = \text{EE}_{\text{opt}} - \text{EE}_{\text{rnd}},$$

where EE_{opt} is the energy efficiency under the learned RIS phase policy derived in Section V, using the formulation in Eq. (28), and EE_{rnd} is the corresponding metric when RIS phases are drawn randomly from a uniform distribution over $[0, 2\pi)^N$. This difference quantifies the additional number of information bits delivered per Joule of system energy due solely to the intelligent RIS control. The figure reveals several important trends that clarify the role of RIS optimization in multi-satellite networks. For a fixed RIS size N, the EE gain ΔEE decreases as the number of cooperating satellites M increases. With few satellites, random RIS phases cause strong signal misalignment and a weak effective channel, so the baseline EE is low; optimizing the RIS phases can then produce a large improvement in the composite channel gain and hence a large ΔEE. As M grows, the baseline becomes more robust because the received power increases through the superposition of multiple independent direct and RIS-reflected links. Consequently, the gap between random and optimized phase control shrinks, leading to a smaller ΔEE.

For any fixed M, ΔEE increases with N. An optimally configured larger RIS enables more coherent combining of reflected components, strengthening the effective channel and reducing the transmit power required to meet a target rate. In contrast, with random phases the RIS contributions add largely non-coherently, so the received-power increase with N is much weaker. This difference in scaling is captured by ΔEE and underscores the value of intelligent phase control in multi-satellite settings.

A larger ΔEE is therefore desirable: it means the optimized RIS can achieve the same throughput with less energy, or higher throughput under the same energy budget. For example, at $M = 2$ and $N = 512$, ΔEE can reach 0.38 bit/J. When $M > 10$, the marginal benefit diminishes; ΔEE drops below 0.07 bit/J even for the largest N, indicating that spatial diversity from multiple satellites already provides high baseline efficiency and reduces reliance on RIS phase alignment.

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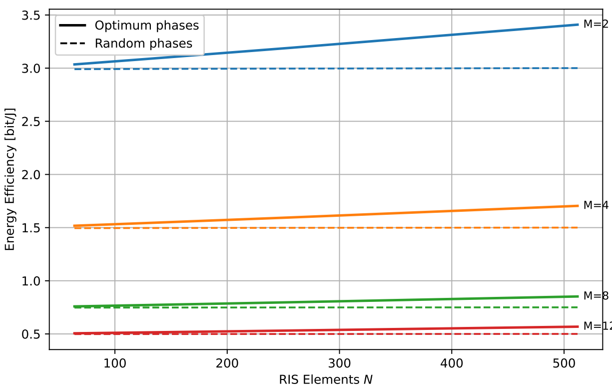

Fig. 4. EE as a function of the number of RIS elements N for different numbers of cooperating satellites M. Solid lines represent optimal RIS phase control, while dashed lines correspond to random phase setting.

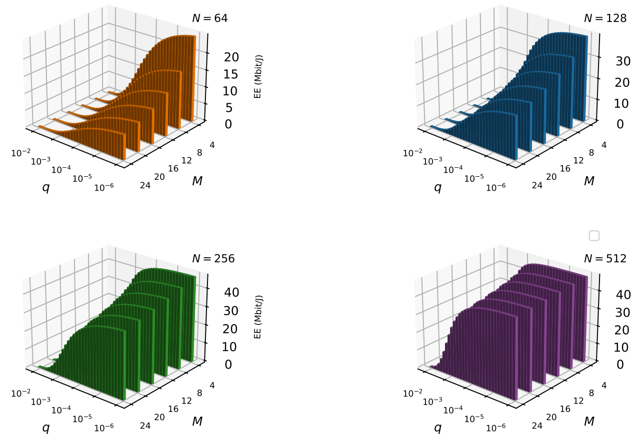

Fig. 6. Instantaneous EE as a function of user mobility noise and satellite count for varying RIS sizes

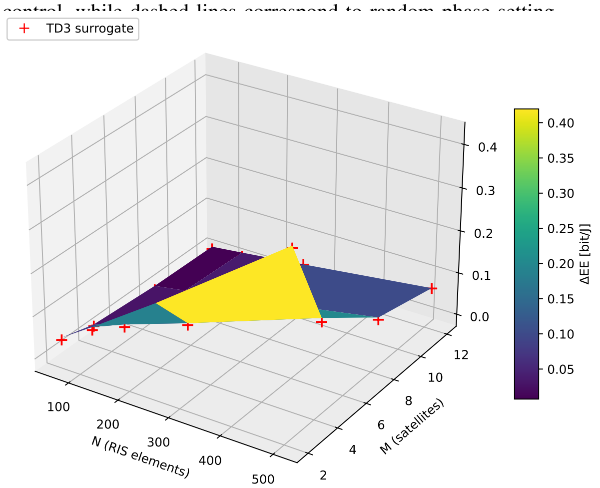

Fig. 5. EE gain surface: Joint effect of RIS aperture and satellite constellation size

Fig. 4 shows the absolute EE (bit J^{-1}) versus RIS size N for $M \in \{2, 4, 8, 12\}$. For each M, the solid curve corresponds to the proposed EKF–TD3 real-time RIS control, whereas the dashed curve is the random-phase baseline with i.i.d. $\phi_n \sim \mathcal{U}[0, 2\pi)$.

With EKF–TD3 (solid curves), EKF-based look-ahead prediction and TD3 phase optimization enable coherent combining, which increases the effective channel gain $|\hat{h}_{\text{eff},m}|^2$ and reduces the required transmit power $P_m^{\min} = \gamma_{\text{req}}\sigma^2/|\hat{h}_{\text{eff},m}|^2$; consequently, EE increases rapidly with N. In contrast, under random phases (dashed curves) the reflected components add incoherently, often causing destructive interference and a loss of array gain, so EE improves only modestly with N.

Fig. 5 illustrates the analytic EE gain surface, ΔEE, defined as the difference in EE between the optimal RIS phase configuration and a random-phase baseline, plotted as a function of the RIS size N and the number M of satellites. The results reveal that ΔEE increases rapidly with the size of the RIS and the satellite count initially, reflecting the significant benefits of coherent phase alignment and spatial diversity. However, as N and M grow larger, the gain plateaus and can even decline. This diminishing return occurs because, with many satellites, the random-phase baseline itself improves through increased diversity, thereby narrowing the advantage of RIS optimization and increasing circuit power consumption. Thus, ΔEE highlights the practical limits of hardware scaling.

Superimposed on this surface, red crosses mark the TD3-based DRL evaluations, which closely match the analytical predictions. This strong agreement demonstrates that the proposed learning-based RIS control achieves near-optimal energy efficiency across a wide range of system parameters.

Fig. 6 presents the instantaneous EE for different RIS sizes ($N = 64, 128, 256$) as a function of the mobility process noise $q \in [10^{-6}, 10^{-2}]$ and the number of active satellites $M \in \{4, 8, \ldots, 24\}$. For all N, EE decreases as q increases, consistent with the state-evolution model in Eq. (12). Larger process noise induces higher prediction error at the planning horizon $t + \hbar$, degrading phase alignment, reducing composite channel gain, and increasing the transmit power required to satisfy the target rate.

EE also declines with increasing M under nonzero prediction uncertainty. Although more satellites provide additional signal paths, the total radiated and circuit power scales approximately linearly with M, while imperfect phase alignment limits coherent gain. Consequently, the required transmit power per satellite does not scale ideally, leading to reduced EE. In the limiting case $q \to 0$, mobility prediction becomes exact, coherent combining is preserved, and the required transmit power per satellite scales approximately as $1/M$ (cf. Eq. (32)), so EE remains stable or may improve with M. This ideal behavior is not achievable under realistic prediction errors.

Comparing the subplots confirms the benefit of increasing the RIS size. As N grows, quadratic array gain dominates the linear increase in circuit power, yielding higher EE. However, for very large N, the circuit-power term eventually offsets the incremental array gain, causing EE saturation and diminishing returns. This highlights the need to optimize N rather than increasing it indefinitely.

These results validate the advantages of the proposed mobility-aware RIS control framework: by integrating accurate user motion prediction (as formulated in Eq. (12)) with scalable RIS deployment, the system achieves high EE even under

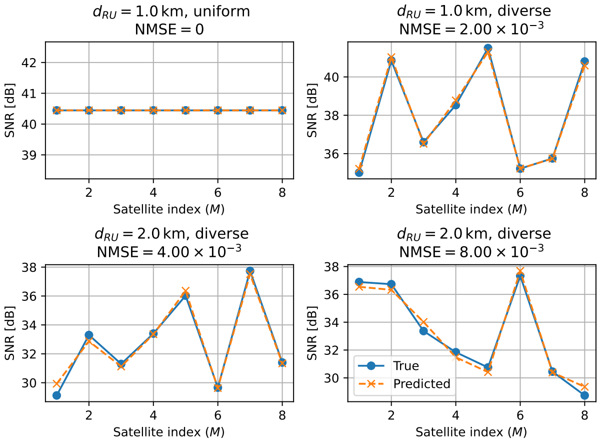

Fig. 7. Predicted and True SNR for RIS-assisted downlink with $N = 128$ elements under varying link conditions and calibration thresholds

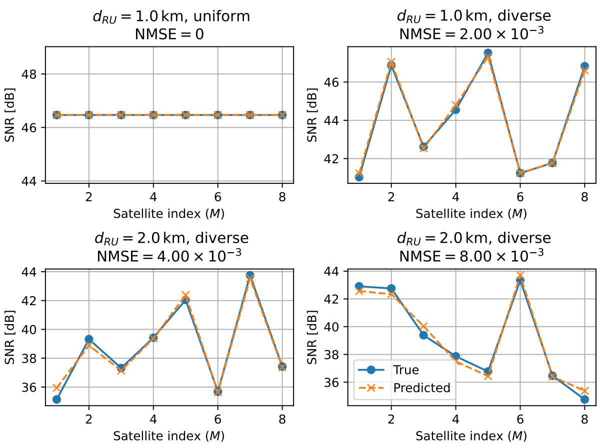

Fig. 8. SNR prediction versus ground Truth for RIS with $N = 256$ elements across channel scenarios

mobility and multi-satellite diversity. This demonstrates the practical energy-saving potential of the approach, particularly in scenarios involving moderate or predictable user motion and an optimized RIS design. The process noise parameter q in Eq. (12) directly characterizes the level of mobility-induced prediction inaccuracy; lower values of q correspond to more accurate tracking and forecasting of the user's position. For example, at low mobility noise ($q = 10^{-6}$), which reflects highly accurate user position prediction, the maximum EE increases by 59% from $N = 64$ to $N = 128$, by 29% from $N = 128$ to $N = 256$, and by 4.4% from $N = 256$ to $N = 512$. On average, this yields a 31% increase in EE per doubling of RIS size, with diminishing returns for very large apertures due to the growing circuit power overhead and the eventual saturation of array gain. These findings highlight the importance of precise user motion prediction for realizing the full energy efficiency benefits of large-scale RIS deployment.

Choice of Planning Horizon and Sensitivity Analysis

The planning horizon $\hbar$ determines how far ahead the EKF–TD3 controller predicts when optimizing RIS phases. A longer horizon provides greater foresight for proactive beamforming but increases EKF prediction error, potentially degrading phase alignment and energy efficiency. To quantify this trade-off, we normalize $\hbar$ by the channel coherence time T_c and define $\eta = \hbar/T_c$. In our baseline setting, $\eta \approx 0.3$, corresponding to a regime where the channel remains strongly correlated.

To evaluate sensitivity to the planning horizon, we swept η over a range of values, including $\eta = 0$ (reactive control). Energy efficiency increases from $\eta = 0$ to moderate values around 0.2–0.4, due to improved short-term prediction. For larger η, the SNR-prediction NMSE grows and energy efficiency declines as prediction errors accumulate. These results confirm that the framework is robust to moderate horizon variations and that the chosen baseline ($\eta \approx 0.3$) lies near the optimal trade-off between foresight and prediction accuracy.

Having established the role of the planning horizon, we next evaluate the accuracy of SNR prediction and its calibration, which underpin the reported energy-efficiency gains.

Figs 7 and 8 jointly illustrate the effectiveness and accuracy of the proposed low-complexity linear calibration method for SNR prediction under varying system conditions and RIS element configurations. In particular, these figures empirically validate the theoretical RIS array gain described by Eq. (9), highlighting the roles of the RIS size N, the element efficiency η_{RIS}, and the phase quantization loss $L_q(b)$.

Fig. 7 considers an RIS comprising $N = 128$ reflecting elements that serve as a baseline reference. It demonstrates four representative scenarios that combine two distinct user-RIS distances, ($d_{\mathrm{RU}} \in \{1.0, 2.0\}\, km$), and two atmospheric fading models: uniform and diverse. The SNR achieved under these scenarios ranges approximately within a 30% window, reflecting realistic channel variability. Crucially, the proposed calibration method achieves highly accurate SNR predictions, with the estimated values closely tracking the ground truth within the explicitly designed NMSE thresholds $0, 2, 4, 8 \times 10^{-3}$. These NMSE levels are selected to represent a realistic spectrum of prediction quality, ranging from ideal (zero error) to moderate levels of estimation noise typically observed in EKF-based mobility and channel tracking for LEO satellite systems. By demonstrating robust performance across this range, the results confirm that the calibration procedure can reliably adapt to both deterministic path loss and stochastic, weather-induced fluctuations, even under non-negligible prediction uncertainties.

In Fig. 8, the RIS size is increased to $N = 256$. Consistent with the RIS gain in Eq. (9), doubling the element count approximately elevates the observed SNR across all scenarios by 15% on the dB scale (around a 6 dB increase), thereby improving overall performance. Despite this substantial increase, the previously established NMSE constraints are maintained across the same four scenarios. The affine estimator inherently incorporates this RIS gain shift, consistently meeting the previously established NMSE constraints across the same four scenarios.

The comparative analysis of Figs 7 and 8 highlights the central enhancement provided by the proposed approach: as the RIS size grows, predictable and stable SNR improvements are achieved without introducing additional computational complexity or estimator retraining. Consequently, the presented calibration method not only guaranties accuracy under

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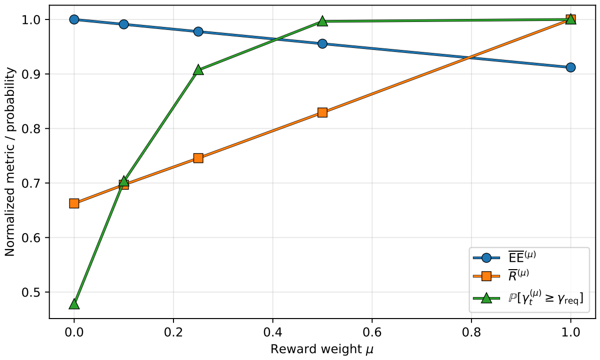

Fig. 9. Sensitivity of normalized energy efficiency $\overline{\mathrm{EE}}^{(\mu)}$, normalized throughput $\overline{R}^{(\mu)}$, and coverage probability $\mathbb{P}[\gamma_t^{(\mu)} \geq \gamma_{\text{req}}]$ to the reward weight μ for $(M, N) = (8, 256)$.

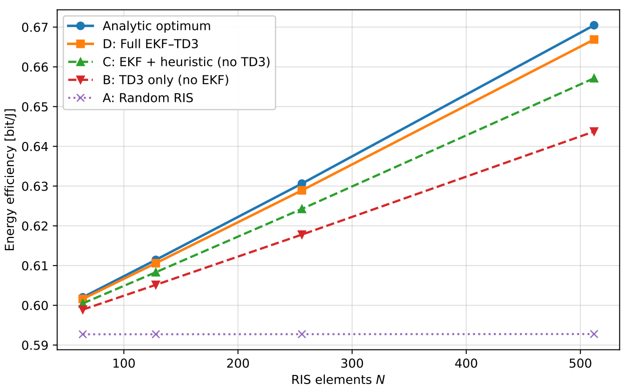

Fig. 10. Ablation study of the EKF–TD3 framework: EE versus RIS aperture N for $M = 8$ satellites, comparing Schemes A–D and the analytical optimum.

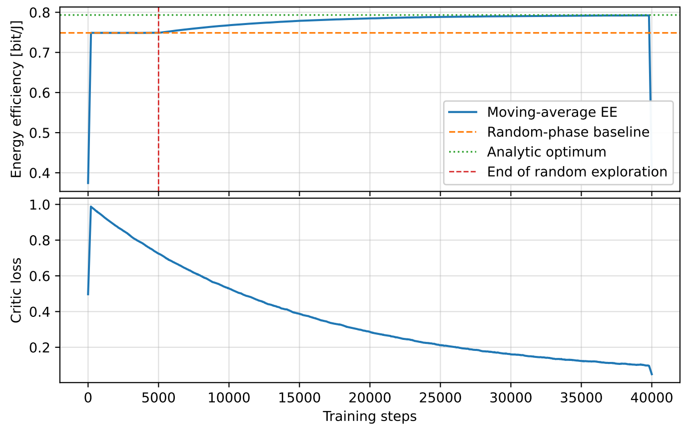

Fig. 11. Convergence behaviour of the TD3-based RIS controller with $(M, N) = (4, 256)$: moving-average energy efficiency compared with the random-phase baseline and the analytic optimum (top), and critic loss (bottom).

diverse channel conditions but also demonstrates scalability with RIS dimensioning, thereby validating its applicability and practical advantages for energy-efficient RIS-assisted satellite communication systems.

To evaluate the robustness of the proposed DRL framework with respect to different optimization objectives, Fig. 9 illustrates the impact of the reward weight μ in (35) on the learned TD3 policy. For each $\mu \in \{0, 0.1, 0.25, 0.5, 1.0\}$, an independent agent is trained, and three metrics are collected over many Monte Carlo realizations for the representative configuration $(M, N) = (8, 256)$: (i) the long-term average energy efficiency $\mathrm{EE}^{(\mu)}$ (bit J^{-1}), (ii) the average user throughput $R^{(\mu)}$, and (iii) the coverage probability $\mathbb{P}[\gamma_t^{(\mu)} \geq \gamma_{\text{req}}]$. For clarity, the figure reports the normalized quantities $\overline{\mathrm{EE}}^{(\mu)} = \mathrm{EE}^{(\mu)} / \max_\mu \mathrm{EE}^{(\mu)}$ and $\overline{R}^{(\mu)} = R^{(\mu)} / \max_\mu R^{(\mu)}$, together with the coverage probability, as functions of μ. The curves exhibit a smooth trade-off: small positive values of μ markedly improve throughput and coverage while incurring only a modest reduction in energy efficiency, whereas larger μ values yield diminishing throughput gains and nearly saturated coverage at the expense of a gradual decrease in $\mathrm{EE}^{(\mu)}$.

The preceding results have primarily compared the full EKF–TD3 framework with the random-phase baseline and the analytical upper bound. However, it is also important to understand the individual role of each module within the proposed architecture. To address this point, Fig.10 presents an ablation study that separates the contributions of the EKF-based prediction module and the TD3-based RIS controller. For a representative configuration with $M = 8$ satellites, the figure shows the energy efficiency as a function of the RIS aperture $N \in \{64, 128, 256, 512\}$ for four schemes: (A) random RIS phases (no EKF, no TD3), (B) TD3-only control without prediction, (C) EKF prediction combined with a simple deterministic beam-steering rule (no TD3), and (D) the proposed full EKF–TD3 framework. The analytical optimum, corresponding to perfect phase alignment, is also included as an upper bound. Across all RIS sizes, the curves exhibit the ordering

$$\text{Analytical optimum} > \text{D (full EKF–TD3)} > \\ \text{C (EKF + heuristic)} > \text{B (TD3 only)} > \text{A (random RIS)},$$

and all schemes benefit from increasing N. The improvement from Scheme A to Scheme B highlights the gain obtained by learning-based RIS control, even without prediction. Scheme C further outperforms Scheme B, showing that EKF-based mobility and channel prediction provide an additional advantage over purely reactive control. Scheme D remains consistently closest to the analytical upper bound, indicating that combining EKF prediction with TD3-based optimization is necessary to achieve the near-optimal energy efficiency reported in the main results.

To illustrate the convergence of the proposed learning-based controller, Fig. 11 shows the evolution of the moving-average energy efficiency and the critic loss versus the training steps for a representative configuration with $M = 4$ satellites and $N = 256$ RIS elements. The initial exploration phase (first 5×10^3 steps) uses purely random actions, and the resulting energy efficiency remains close to the random-phase baseline. Once the TD3 updates are activated, the moving-average EE rises smoothly and eventually saturates near the analytic upper bound derived in Section III-A, while the critic loss decreases steadily to a small value. This behaviour indicates that the actor–critic pair has reached a stable policy whose performance consistently outperforms the random-phase baseline and approaches the theoretical optimum. All numerical results reported in Section IV are obtained from this converged training regime.

Computational Complexity

To complement the performance results in the previous subsections, we briefly discuss the computational complexity and real-time feasibility of the proposed EKF–TD3 framework. The per-decision cost can be decomposed into three parts: the EKF-based mobility prediction, the SNR calibration step, and the forward pass of the TD3 actor network.

First, the EKF operates on a four-dimensional state vector $\mathbf{x}(t) = [x(t), \dot{x}(t), y(t), \dot{y}(t)]^\top$ and a measurement vector of dimension $3M$ (azimuth, elevation, and time-of-flight from each of the M visible satellites). The prediction and update recursions in (15)–(16) involve matrix multiplications with small 4×4 and $4 \times 3M$ matrices, so the per-step complexity scales linearly with the number of satellites, i.e., on the order of $\mathcal{O}(M)$ floating-point operations with a very small constant factor.

Second, the affine SNR-calibration rule introduced in Section V consists of a single linear transformation with a fixed set of coefficients. Its complexity per decision is therefore $\mathcal{O}(1)$ and negligible compared with the prediction and control components.

Third, the TD3 agent selects the RIS phase vector by evaluating the actor network $\pi_\psi(\mathbf{s}_t)$ once per decision interval. In our implementation, the actor is a fully connected network with two hidden layers of moderate width. Let d_s and d_a denote the state and action dimensions, respectively, with $d_a = N$ and d_s containing a low-dimensional representation of the predicted position, composite channels, and current phase configuration. The number of real multiplications in a forward pass is then approximately proportional to $d_s H + H^2 + H d_a$, where H is the number of hidden neurons. For the considered configurations ($N \leq 512$, $M \leq 12$), this results in a few $\times 10^4$ floating-point operations per decision, scaling roughly linearly with N and M.

Importantly, only the actor forward pass is required during on-line operation. The critic networks and replay-buffer updates appear only in the training loop (Algorithm 2) and can be executed offline or on a more powerful ground processor. The on-board real-time decision making therefore reduces to one EKF update, one affine SNR calibration, and one small feedforward neural-network evaluation per time slot. The corresponding order-of-growth per-decision costs are summarized in Table II, and they remain well within typical processing capabilities of modern LEO satellite or RIS controller hardware.

Finally, we briefly comment on scalability to larger multi-beam and multi-user settings. In such extensions, the state vector can be augmented with compact per-beam or per-user summaries (e.g., effective channel gains, required SINR targets), so that its dimension grows approximately linearly with the number of beams and users rather than exponentially. The actor forward-pass complexity remains proportional to $d_s H + H^2 + H d_a$, where d_s and d_a are the state and action dimensions and H is the hidden-layer width. Likewise, the memory footprint of the replay buffer scales as $\mathcal{O}(N_{\text{buf}}(d_s + d_a))$. Thus, both time and memory requirements grow polynomially in the number of beams, users, satellites, and RIS elements, which supports the scalability of the proposed framework to larger STIN deployments.

TABLE II
ORDER-OF-GROWTH PER-DECISION COMPLEXITY OF THE MAIN COMPONENTS OF THE PROPOSED FRAMEWORK.

Component	Complexity per step
EKF prediction and update	$\mathcal{O}(M)$
SNR calibration	$\mathcal{O}(1)$
TD3 actor forward pass	$\mathcal{O}(N + M)$ (for fixed hidden size H)

VII. Conclusion

The study has established a unified, mobility-aware framework for EE downlink communication in RIS-assisted LEO satellite networks by integrating EKF mobility prediction, a single-step affine SNR calibration rule, and DRL optimisation of RIS phase shifts. The calibration procedure was demonstrated, both analytically and through extensive simulations, to enforce any prescribed NMSE target. In the present evaluation, the prediction error was maintained below 8×10^{-3} across all evaluated scenarios, satellite multiplicity, and RIS aperture, thus demonstrating reliable channel tracking under realistic mobility and fading. Scalability with the RIS surface size was confirmed by doubling the element count from 128 to 256, which yielded the theoretically expected SNR increase of approximately 6 dB without requiring estimator retraining. This deterministic gain translated directly into transmit power savings that exceeded 50% on a per-satellite basis in typical operating regimes and produced a network-wide power reduction of up to 12% relative to conventional, non-predictive baselines. When the RIS phases were optimized through the proposed TD3 agent, the realized EE was between 93% and 97% of the analytical optimum, with absolute values ranging from 0.5 to 3.4 bits J^{-1} across the tested (M, N) grid; the incremental gain attributable to learning-based RIS control reached 0.38 bits J^{-1} for moderate satellite diversity and apertures of 512 elements.

These results confirm that accurate SNR prediction, low-overhead calibration, and data-driven RIS control can be combined into a practical, hardware-aware architecture that consistently delivers high energy efficiency, substantial reductions in transmit power, and near-optimal performance across a broad spectrum of mobility and channel conditions. The framework, therefore, provides a viable pathway toward scalable, high-throughput, and energy-constrained LEO communications, enabled by intelligent reconfigurable surfaces.

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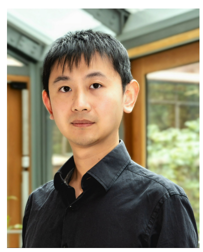
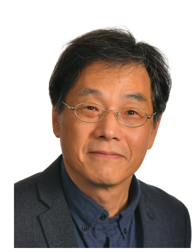
Muhammad I. Khalil (M'14) is a researcher in the School of Electrical and Electronic Engineering at RMIT University in Melbourne, Australia. He received the B.Sc. degree in Electrical Engineering from the University of Mosul in 2000, the M.Sc. degree in Electrical Engineering from the University of Baghdad, Iraq, in 2006, and the Ph.D. degree in Electrical and Electronic Engineering from the University of Auckland, New Zealand, in 2017. In 2021, he served as a Postdoctoral Research Fellow at the University of New South Wales in Sydney, Australia. His research interests encompass next-generation mobile and wireless networks, vehicular communication systems, satellite communications, and the design of intelligent converged communication infrastructures.



Ke Wang Ke Wang received the Ph.D. degree in Electrical and Electronic Engineering from the University of Melbourne, Australia, in 2014. He is currently an Associate Professor in the School of Engineering at the Royal Melbourne Institute of Technology (RMIT University), Australia. He was previously an Assistant Professor at Stanford University, USA, and an ARC DECRA Fellow at the University of Melbourne. His current research interests include optical and wireless communication technologies and their convergence, terahertz communications, satellite communications and networks, machine learning algorithms and hardware acceleration, and silicon photonic integration.



Jinho Choi (Fellow, IEEE) received his B.E. (magna cum laude) in Electronics Engineering from Sogang University, Seoul, in 1989, and his M.S.E. and Ph.D. degrees in Electrical Engineering from KAIST in 1991 and 1994, respectively. He is currently a Professor in the School of Electrical and Mechanical Engineering at the University of Adelaide, Australia. His research interests include the Internet of Things (IoT), wireless communications, and statistical signal processing. He is the author of three books and a recipient of several best paper awards, including the 1999 EURASIP Best Paper Award in Signal Processing. Prof. Choi was elected IEEE Fellow for his contributions to wireless communications and signal processing and has been ranked among Stanford University's World's Top 2% Scientists since 2020. He is Senior Editor of IEEE Wireless Communications Letters and Associate Editor of IEEE Transactions on Mobile Computing, and previously served as Division Editor of the Journal of Communications and Networks and as Associate Editor/Editor for IEEE Transactions on Communications, IEEE Communications Letters, IEEE Transactions on Vehicular Technology, and the ETRI Journal.